

University of Minho
School of Engineering



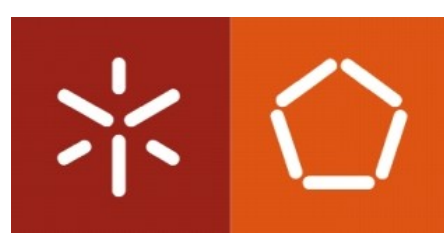
Cryptography & Network Security

Bachelor in Telecommunications and Informatics Engineering

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2024/2025



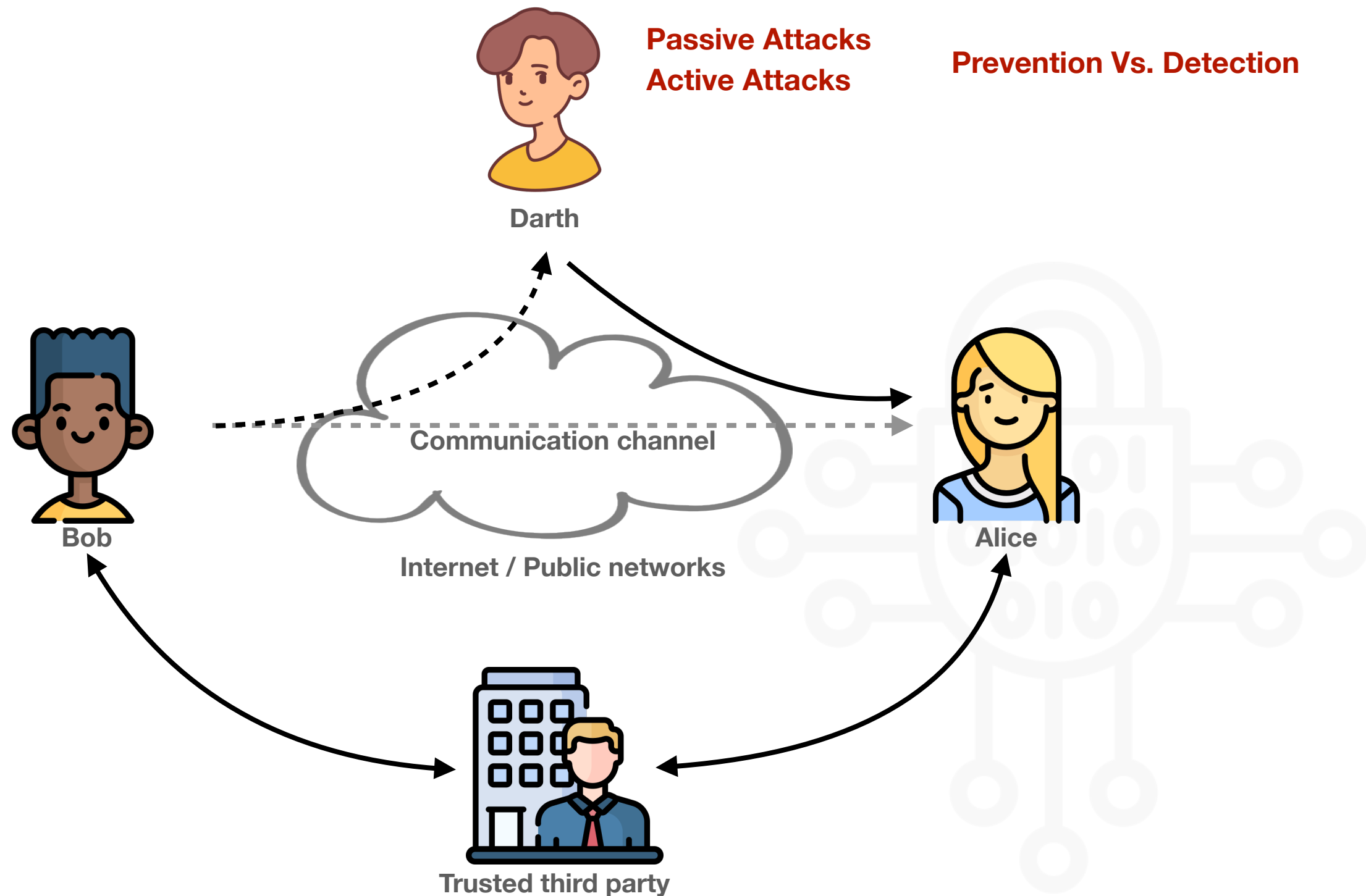
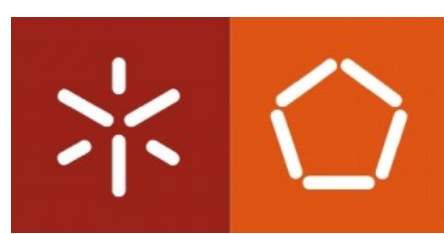
Part II

Symmetric Encryption

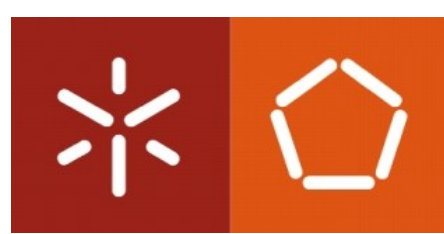


Principles

Communication Model



Principles



What is Cryptography?

The discipline that embodies the principles, means, and methods for the transformation of data to hide their semantic content, prevent their unauthorised use, or prevent undetected modification.

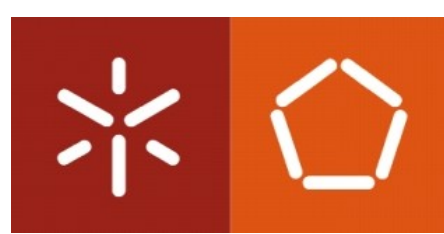
NIST SP 1800-21B

It aims at protecting security properties

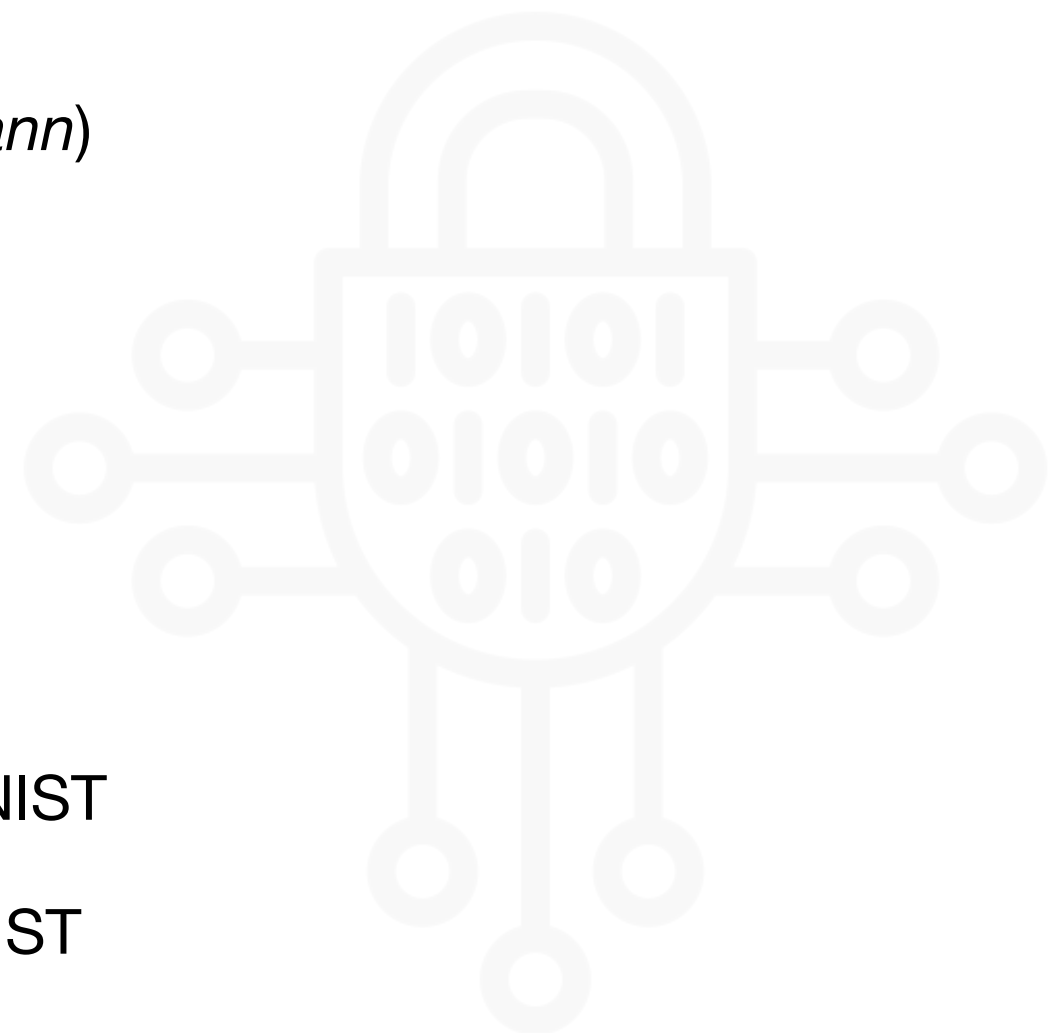
- Confidentiality
- Integrity
- Availability
- Authenticity
- Non-repudiation



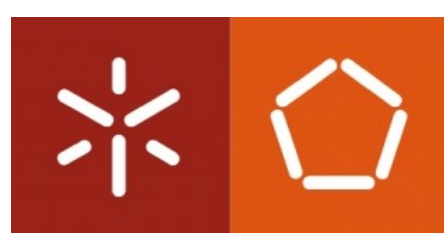
Principles



- Summary of the History of Modern Cryptography
 - 1948-1949: Cryptography was initiated as a scientific area after Claude Shannon's development of the Information Theory that allows formalising notions of security.
 - 1970-1977: Data Encryption Standard (DES).
 - 1976: Asymmetric Cryptography (*Diffie & Hellmann*)
 - 1978: *Rivest, Shamir, & Adelman* (RSA)
 - 1985: *El Gamal*.
 - 1995: Digital Signature Algorithm (DSA)
 - 2001: Advanced Encryption Standard (AES)
 - 2016: Report on Post-Quantum Cryptography, NIST
 - 2022: Selection of Post-Quantum Algorithms, NIST



Principles



Plaintext: the original message or data fed into a cryptographic algorithm as input.

Encryption algorithm: a set of mathematically expressed rules for rendering data unintelligible by executing a series of conversions controlled by a key.

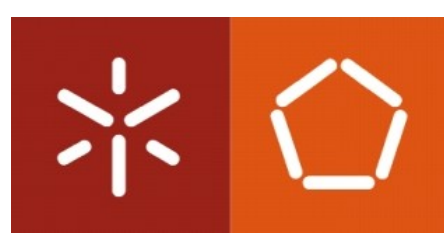
CNSSI 4009-2015

Secret key: an input the encryption algorithm uses to perform the substitutions and transformation in the plaintext.

Ciphertext: the scrambled message produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts.

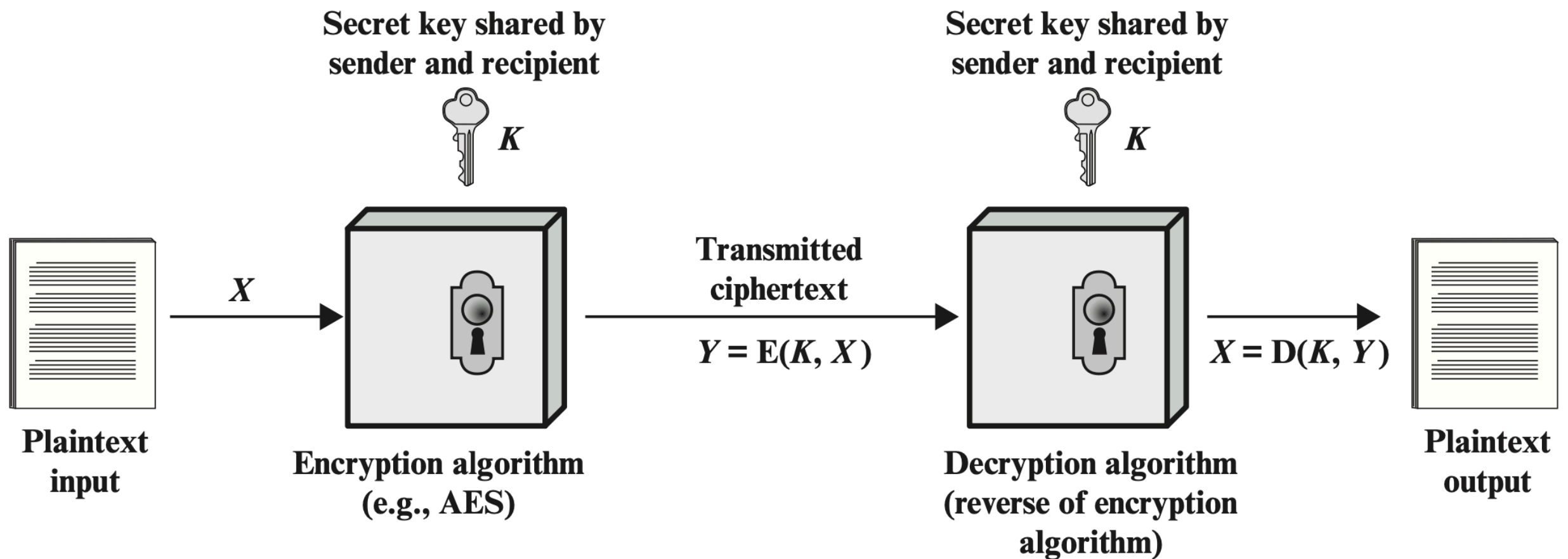
Decryption algorithm: a set of rules that take the ciphertext and the secret key to produce the original plaintext. This is essentially the encryption algorithm run in reverse.

In symmetric cryptography, the encryption and decryption algorithms use the same key.



Principles

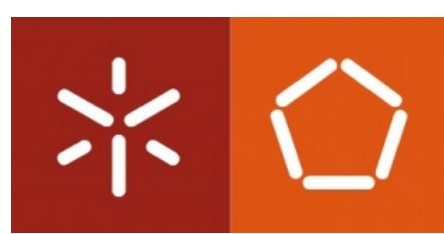
Simplified Model of Symmetric Encryption



Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Principles

Operation



Block Ciphers: the cryptographic algorithm operates on fixed-length blocks of the input (e.g., plaintext) with the resulting output block (e.g., ciphertext) of the same size.

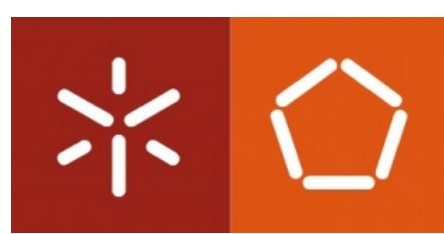
- Common block sizes are 64 and 128 bits

Stream Ciphers: the cryptographic algorithm processes an individual bit, byte, or character of the input (e.g., plaintext) at a time.

Substitution: each plaintext element or group of elements is uniquely replaced by a corresponding ciphertext element or group of elements

Permutation: a sequence of plaintext elements is replaced by a permutation of the sequence without adding or removing any element from the original sequence.

Principles

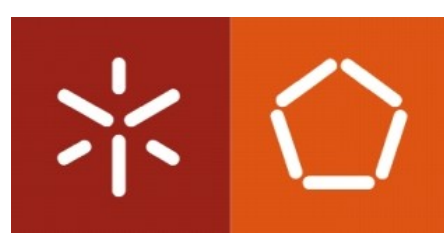


Cryptanalysis: the study of mathematical techniques for defeating cryptographic techniques without an initial knowledge of the used key. This includes looking for errors or weaknesses in the implementation of an algorithm or the algorithm itself.

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Cryptology: the science that deal with cryptography and cryptanalysis.

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Principles

Attacks on Cryptographic Systems

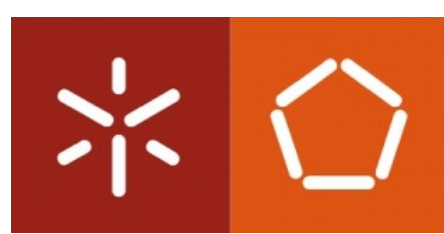
The objective is either to recover the plaintext or the secret key in use.

Cryptanalysis

Rely on the nature of the algorithm plus perhaps some knowledge of the general characteristics of the plaintext or even some sample plaintext-ciphertext pairs. The known characteristics are used to deduce a specific plaintext or the secret key.

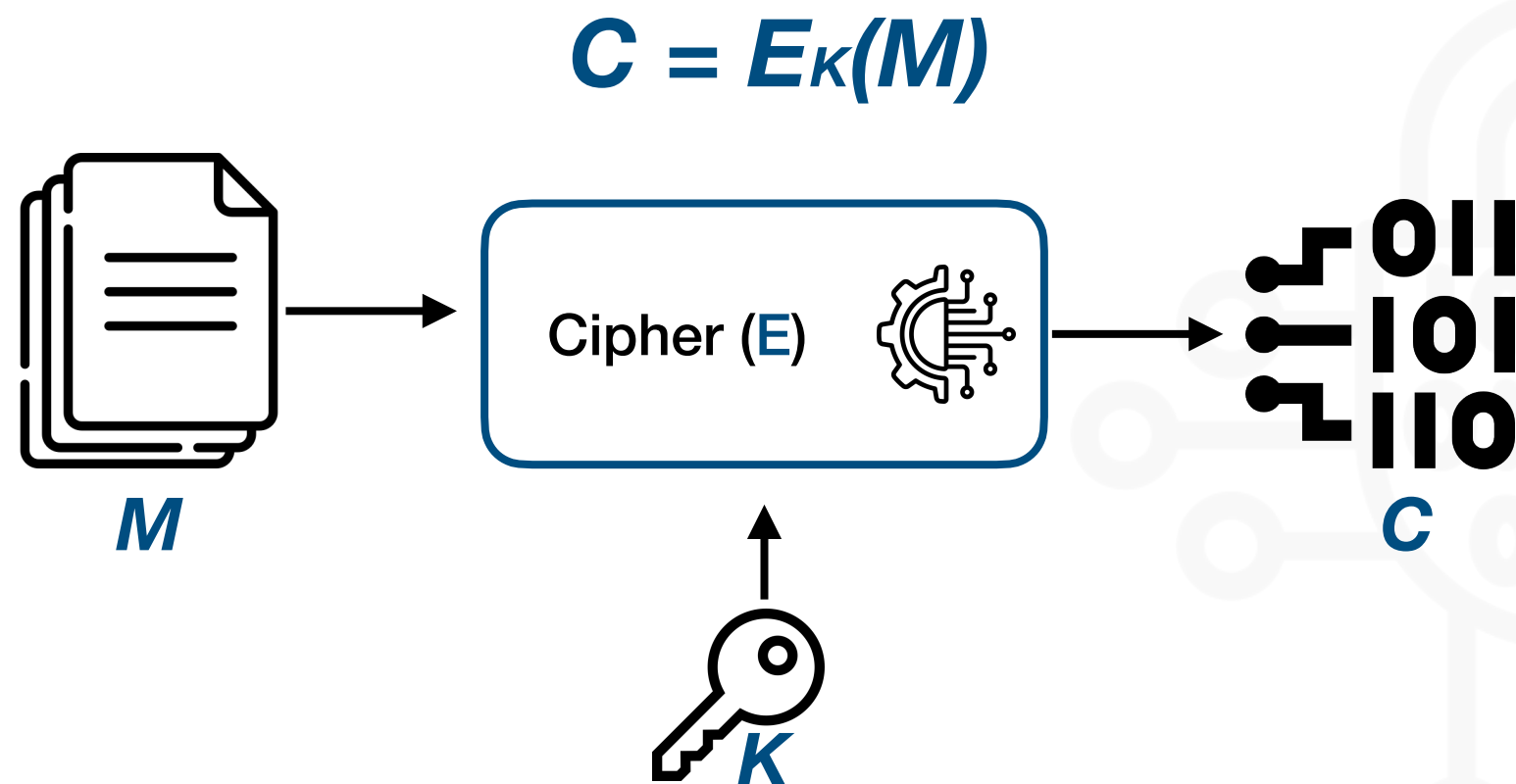
Brute-force attack

The attacker tries every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained. On average, half of all possible keys must be tried to achieve success.

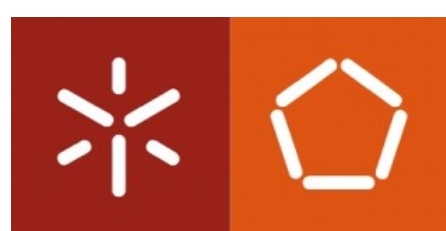


Kerckhoff's Principle

A cryptosystem should be secure even if everything about the system, except the key, is public knowledge.



Its security is ensured by the Secret Key



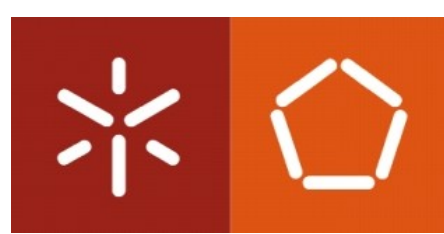
Principles

Cryptanalysis - Types of Attacks

Type of Attack	Known to Cryptanalyst
Ciphertext Only	■ Encryption algorithm ■ Ciphertext
Known Plaintext	■ Encryption algorithm ■ Ciphertext ■ One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	■ Encryption algorithm ■ Ciphertext ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

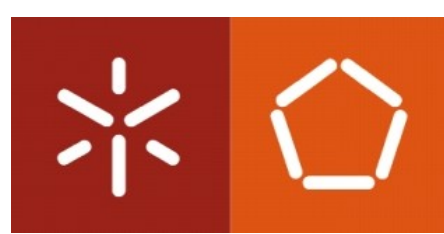
Principles



An encryption scheme is **unconditionally secure** if the ciphertext generated by the scheme does not contain enough information to recover the corresponding plaintext, no matter how much ciphertext is available and how much time an attacker has.

An encryption scheme is **computationally secure** if the ciphertext generated by the scheme meets one or both of the following criteria:

- The cost of breaking the cipher exceeds the value of the encrypted information;
- The time required to break the cipher exceeds the useful lifetime of the information.



Classical Techniques

Caesar Cipher

A substitution cipher allegedly created by Julius Caesar during the Gallic campaign.

It works by replacing each letter of the alphabet with the letter standing three places further down the alphabet.

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Example

Plaintext: attackatnineam

Ciphertext: DWWDFNDWQLQHDP

It is possible to use an arbitrary offset (key).

Plaintext: attackatnineam

Ciphertext (k=6): GZZGIQGZTGTKGS



**26 possible keys.
One of them is insecure.**



Classical Techniques

Caesar Cipher

Algorithm

By assigning a numerical equivalent to each letter, the algorithm can be expressed as follows:

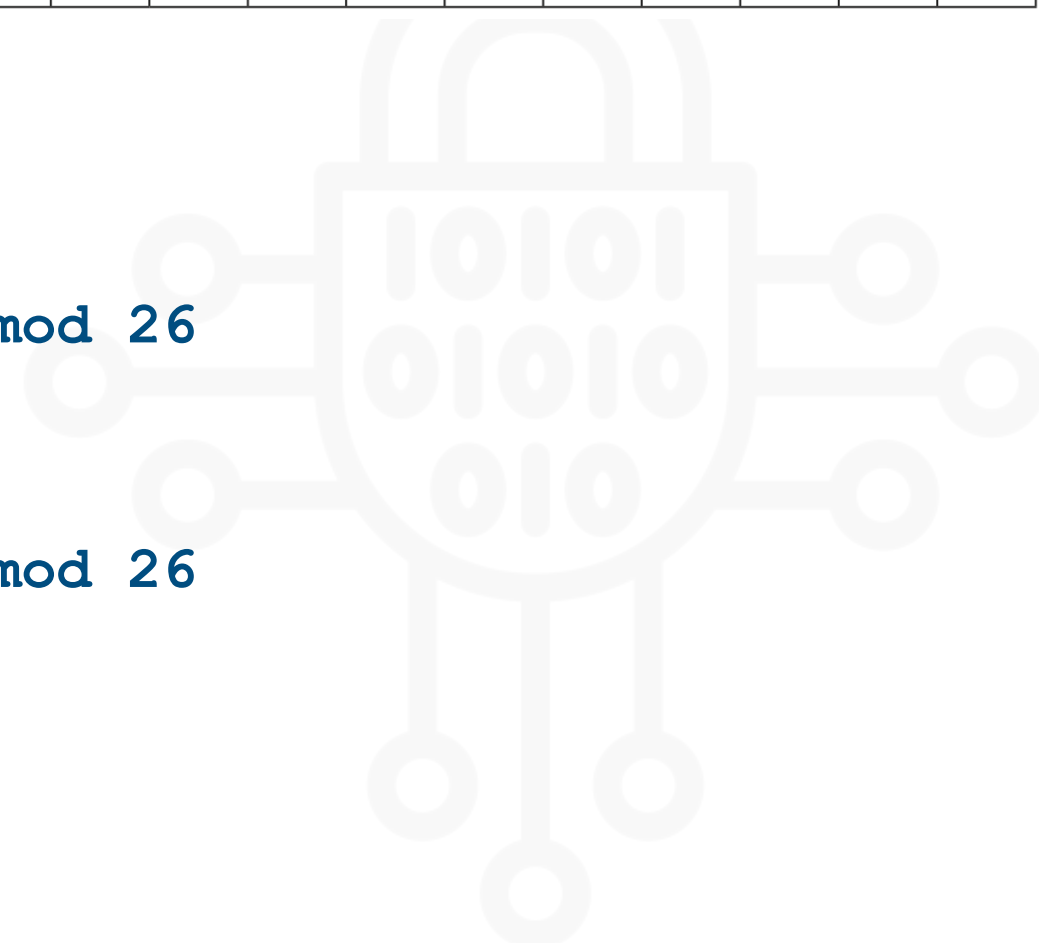
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

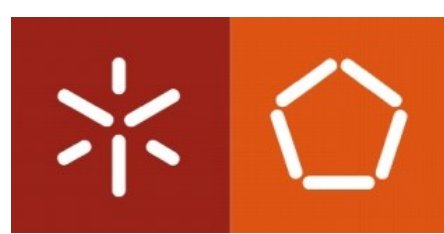
For each plaintext p , substitute the ciphertext letter C :

$$C = E(k, p) = (p + k) \bmod 26$$

The decryption algorithm is:

$$p = D(k, C) = (C - k) \bmod 26$$





Classical Techniques

Caesar Cipher

Cryptanalysis

- The encryption and decryption algorithms are known
- There are only 26 keys to try
- The language of the plaintext is known and easily recognisable

Ciphertext: **XQQXZHXQKFKBXJ**

Key = 1: **YRRYAIYRLGLCYK**

Key = 2: **ZSSZBJZSMHMDZL**

Key = 3: **ATTACKATNINEAM**

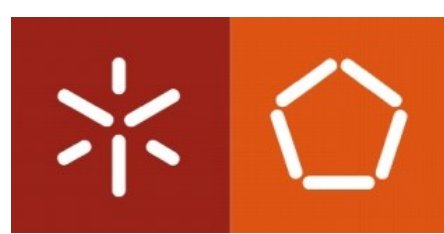
Key = 4: **BUUBDLBUOJOFBN**

...

Key = 24: **VOOVXFVOIDIZVH**

Key = 25: **WPPWYGWPJEJAWI**

Easy brute-force cryptanalysis



Classical Techniques

Caesar Cipher

Cryptanalysis

- The encryption and decryption algorithms are known
- There are only 26 keys to try
- The language of the plaintext is known and easily recognisable

The plaintext output may not be recognizable if the plaintext language is unknown.

Example

Original plaintext: attackatnoonafterthesecondringofthechurchbells

Compressed plaintext: ILIOHNMKKOr3J3RsrPfz007Wfz00



Classical Techniques

Key Space

In most network situations, we can assume that the ciphers are known. What generally makes brute-force cryptanalysis impractical is the use of an algorithm that employs a large key space.

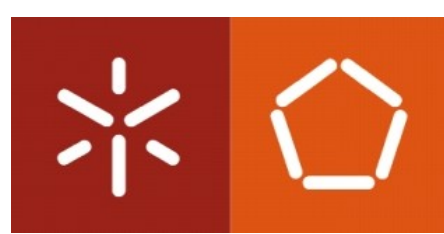
Estimated time for brute-force cryptanalysis

Key size	Time ($1\mu\text{sec}/\text{test}$)	Time ($1\mu\text{sec}/10^6\text{test}$)
32 bit	35.8 min	2.15 msec
40 bit	6.4 days	550 msec
56 bit	1140 years	10 hours
64 bit	500000 years	107 days
128 bit	5×10^{24} years	5×10^{18} years

- Note that increasing the key size by one bit doubles the available key space.
- Currently, 2^{80} is considered to provide an acceptable level of security.

Classical Techniques

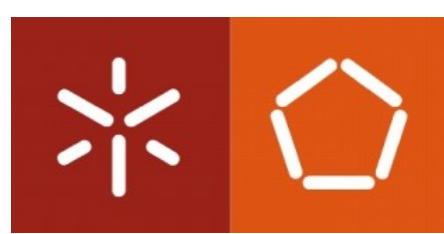
Caesar Cipher



Exercise

Write a program that implements encryption and decryption functions according to the Caesar Cipher.





Classical Techniques

Monoalphabetic Ciphers

- Enhances Caesar cipher by allowing arbitrary substitutions (i.e., *permutations*)
- A single cipher alphabet (mapping from plain alphabet to cipher alphabet) is used per message
- Increases the key space

Example

Consider the finite set of elements $S = \{a, b, c\}$

There are six permutations of S : *abc, acb, bac, bca, cab, cba*

What are the number of possible permutations for the alphabet?

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

In general, there are $n!$ Permutations of a set of n elements

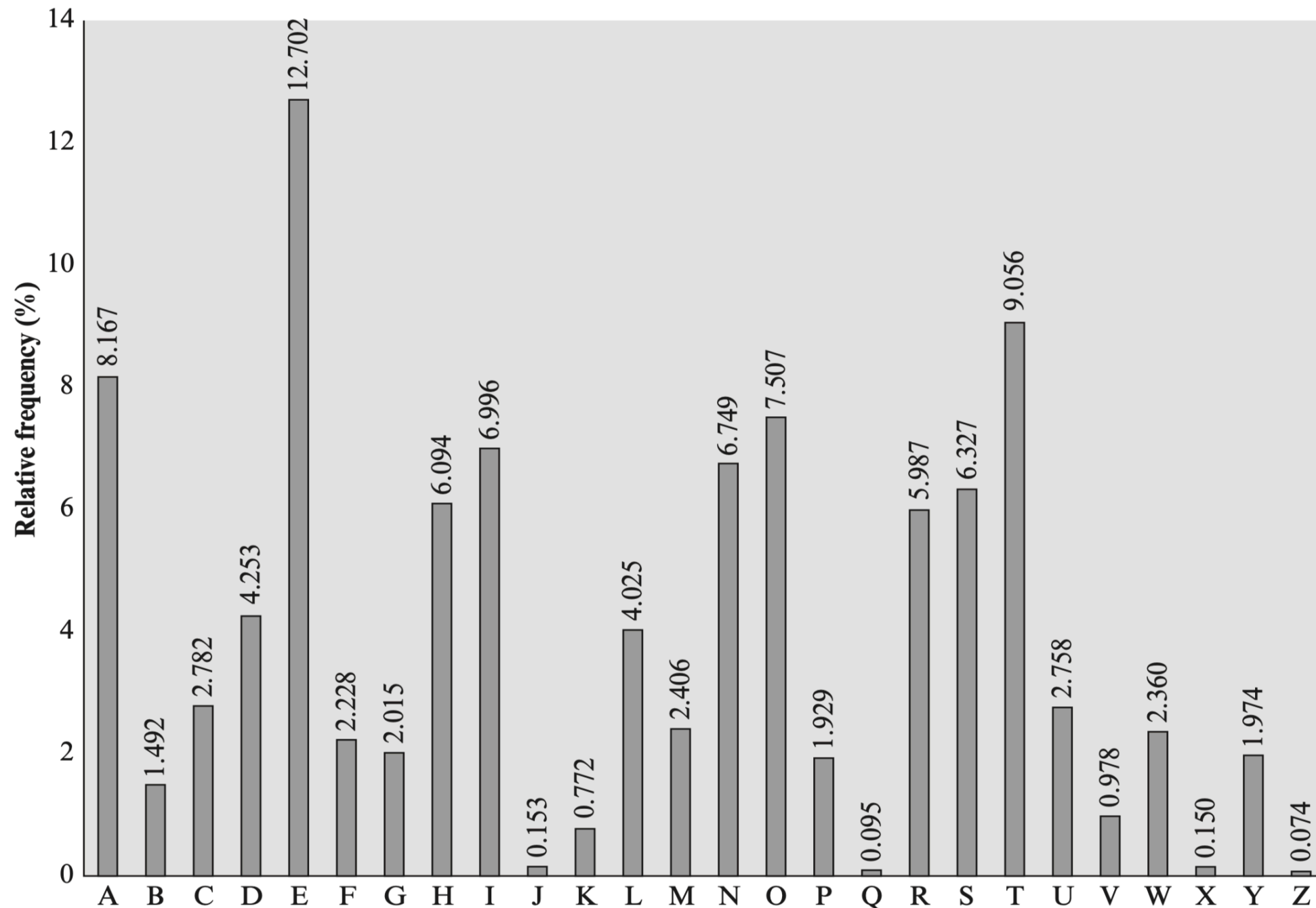
$26! > 4 \times 10^{26} \longrightarrow$ Strong against brute-force cryptanalysis

Classical Techniques

Monoalphabetic Ciphers

Cryptanalysis

Relative Frequency of Letters in English Text



Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice



Classical Techniques

Monoalphabetic Ciphers

Cryptanalysis - Example

Ciphertext

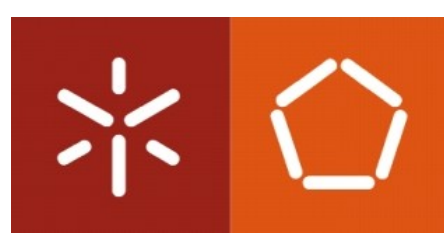
UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAI Z
VUEPHZHMDZSHZOWSFPAPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

Bigrams and trigrams are also useful

The relative frequencies of the letters in the ciphertext

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

Use this to recover the plaintext!



Classical Techniques

Monoalphabetic Ciphers

Cryptanalysis - Result

Ciphertext

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAI Z
VUEPHZHMDZSHZOWSFPAPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

Plaintext

it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow

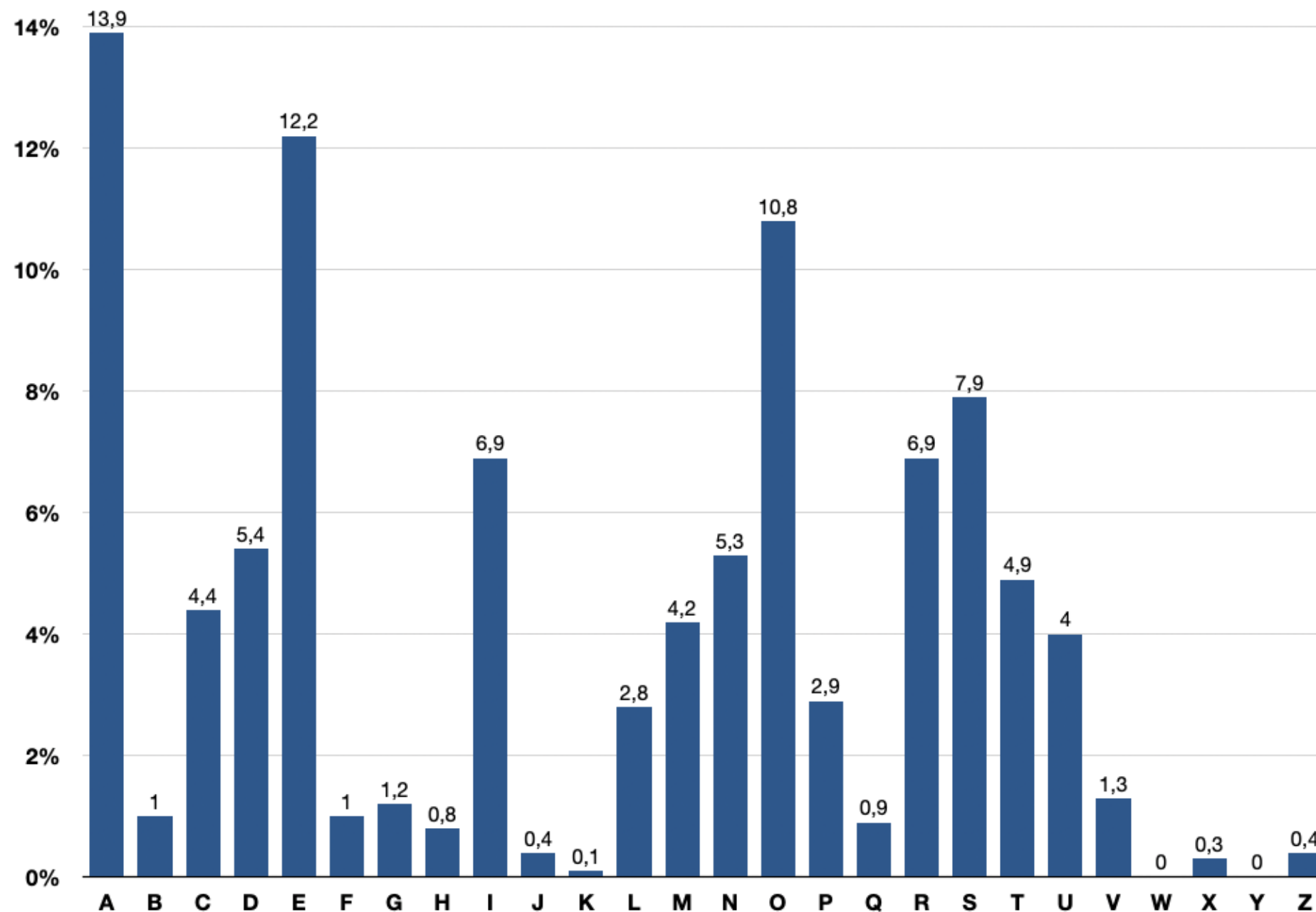


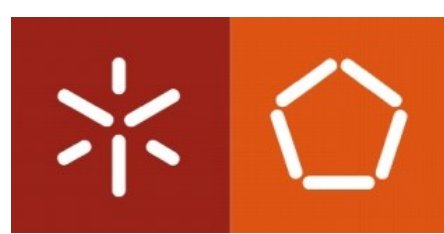
Classical Techniques

Monoalphabetic Ciphers

Cryptanalysis

Relative Frequency of Letters in Portuguese Text





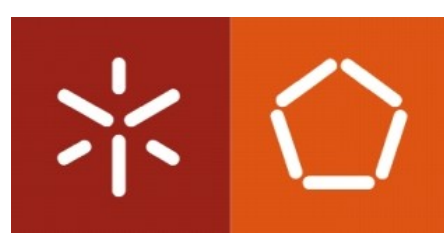
Classical Techniques

Monoalphabetic Ciphers

Exercise

Extend the Caesar Cipher program to support permutations of the alphabet. In practice, what constitutes the secret key?





Classical Techniques

Polyalphabetic Ciphers - Vigenère Cipher

- Introduced by Blaise Vigenère (late 16th century) and known as “*le chaffer indéchiffrable*”.
- It consists of 26 Caesar Ciphers with shifts of through 25.
- It was broken at the end of the 19th century by Charles Babbage and Friedrich Kasiski.

Description

- **The Key is a "phrase" in which each letter determines a substitution.**
- **Each substitution is a simple shift determined by the respective character in the key (e.g., $A=0$, $B=1$, ..., $Z=25$).**
- **The Key size defines the number of substitutions that are used.**



Classical Techniques

Polyalphabetic Ciphers - Vigenère Cipher

Example

Original plaintext: attackatnoonafterthesecondringofthechurchbells

Key: SECRET

Ciphertext: SXVRGDSXPFSGSJVVMZIUVGHFHTZRZGJVYIVZYTTLUWPNJ

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Original plaintext:

a t t a c k a t n o o n

Key:

S E C R E T S E C R E T

Ciphertext:

S X V R G D ...

→ Note that the same input produces different outputs.



Classical Techniques

Polyalphabetic Ciphers - Vigenère Cipher

Algorithm

By assigning a numerical equivalent to each letter, the algorithm can be expressed as follows:

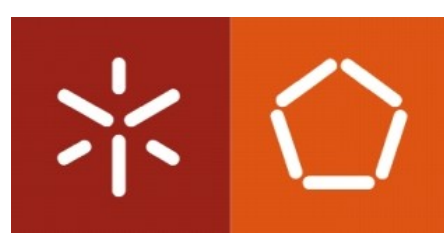
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

For a plaintext $p = p_0, p_1, p_2, \dots, p_{(n-1)}$ and a key $k = k_0, k_1, \dots, k_{(m-1)}$, the general encryption equation for C_i is:

$$C_i = (p_i + k_{i \bmod m}) \bmod 26$$

The decryption generalization is:

$$p_i = (C_i - k_{i \bmod m}) \bmod 26$$



Classical Techniques

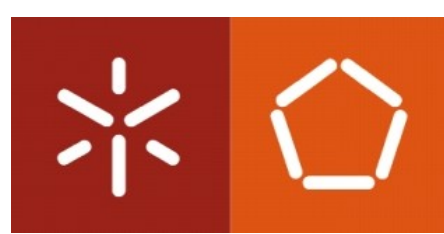
Polyalphabetic Ciphers - Vigenère Cipher

Cryptanalysis

Cipher broken at the end of the 19th century by Charles Babbage and Friedrich Kasiski.

- **Infer probable sizes for the key (e.g., by analyzing the spacing between repeated patterns - Kasiski test);**
- **Break the cryptogram into "slices" (one for each letter of the key);**
- **Attack each of them (Caesar cipher) using frequency analysis.**

Explore the Vigenère Cipher's cryptanalysis in detail at this [link](#).



Classical Techniques

Transposition

Transposition ciphers generate ciphertext by performing some sort of permutation on the plaintext letters.

Example - A simple transposition scheme

Original plaintext: attackatnoon

Key: 4213 (this is the permutation order)

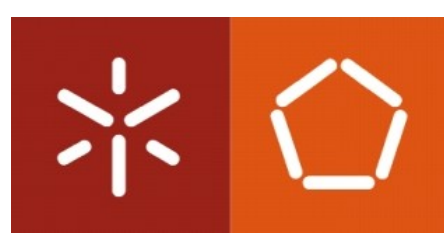
1	2	3	4
a	t	t	a
c	k	a	t
n	o	o	n

← **Read the matrix following the key order**

Ciphertext: ATATTKCANONO

Cryptanalysis

The ciphertext preserves the frequency of the letters from the plaintext.



Classical Techniques

One-Time Pad

An unconditionally secure cipher, that:

- Uses a key as long as the message (*i.e.*, plaintext);
- The key is randomly generated;
- Each key is used to encrypt and decrypt a single message.

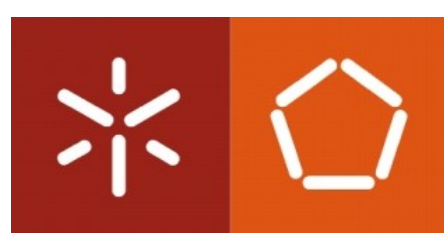
It produces random output that bears no statistical relationship to the plaintext.

It usually operates over a binary alphabet using *XOR* operations.

$$C_i = p_i \text{ XOR } k_i$$

$$p_i = C_i \text{ XOR } k_i$$

In practice, its security is entirely due to the randomness of the key. If the stream of characters (or bits) in the key is truly random, then so are the characters (or bits) in the ciphertext.

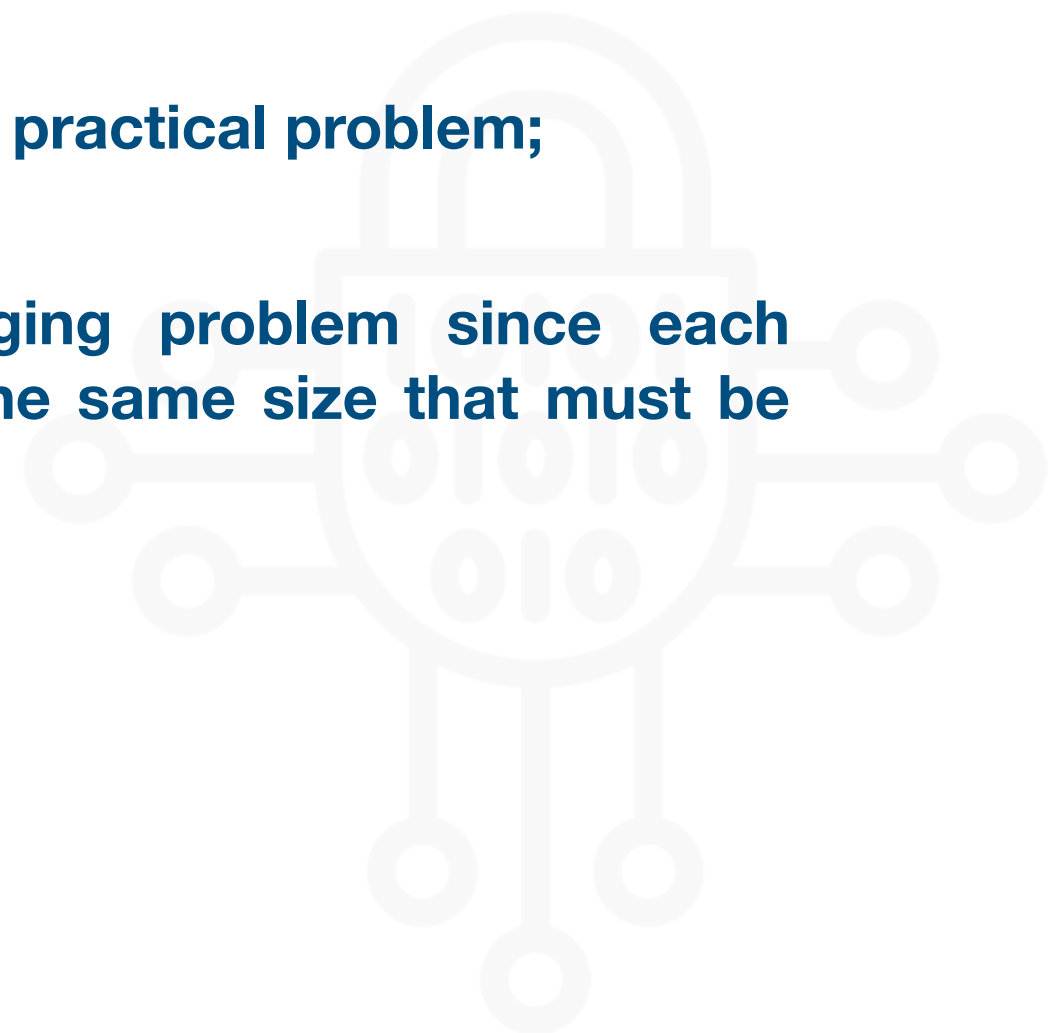


Classical Techniques

One-Time Pad

Despite offering unconditional security, the one-time pad has two fundamental drawbacks:

- **Generating large numbers of random keys is a practical problem;**
- **Key distribution is an even more challenging problem since each message to be sent requires a new key of the same size that must be shared with the receiver.**





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